3rd Homework IR

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Contents

1 Task 1: World Frame and Body Frames

1.1 World Frame {S}

The world frame is defined at the global origin:

$$\text{Origin}_S = (0, 0, 0)$$

1.2 Body Frames

We define fixed body frames for the objects relative to {S}.

Blue Cube (Frame {B}) The frame is located at the center of the blue cube:

$$p_{S\to B} = [17.5,\ 2.5,\ 2.5]^T$$

$$T_{S\to B} = \begin{bmatrix} 1 & 0 & 0 & 17.5 \\ 0 & 1 & 0 & 2.5 \\ 0 & 0 & 1 & 2.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Red Box (Frame {R}) The frame is located at the center of the red box:

$$p_{S\to R} = [10,\ 12.5,\ 10]^T$$

$$T_{S\to R} = \begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 12.5 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 Task 2: Manipulator Base Frame Selection

To simplify the workspace analysis, the manipulator base frame {M} is chosen to coincide with the World Frame {S}.

$$T_{S\to M} = I_{4\times 4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3 Task 3: End-Effector Grasp Strategy & Matrix

3.1 Grasp Position Strategy (Top Grasp)

To successfully grasp the cube, we define the target position for the end-effector (p_E) at the **center of the top surface** of the blue cube. Given that the cube's center is at $x = 17.5$,at $y = 2.5$ cm and at $z = 5.0$ cm.

$$p_E = [17.5,\ 2.5,\ 5]^T$$

We select this top-down approach position because, with this placement the Newton-Raphson inverse kinematics algorithm converges efficiently in just 8 iterations.

Joints	**Min angle (°)**	**Max angle (°)**
Joint 1	-160	160
Joint 2	-70	115
Joint 3	-170	170
Joint 4	-113	75
Joint 5	-170	170
Joint 6	-115	115
Joint 7	-180	180

Table 1: Joint limits (in degrees) for the Elephant myArm manipulator.

3.2 Relative Transformation Matrix ($T_{E \to B}$)

Based on the top grasp strategy defined above, we determine the relative transformation matrix between the end-effector {E} and the cube {B}.

Since the gripper is positioned at the top surface ($z = 5$) and the cube center is at ($z = 2.5$), the cube center is located 2.5 cm **below** the gripper (along the gripper's Z-axis).

$$T_{E \to B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

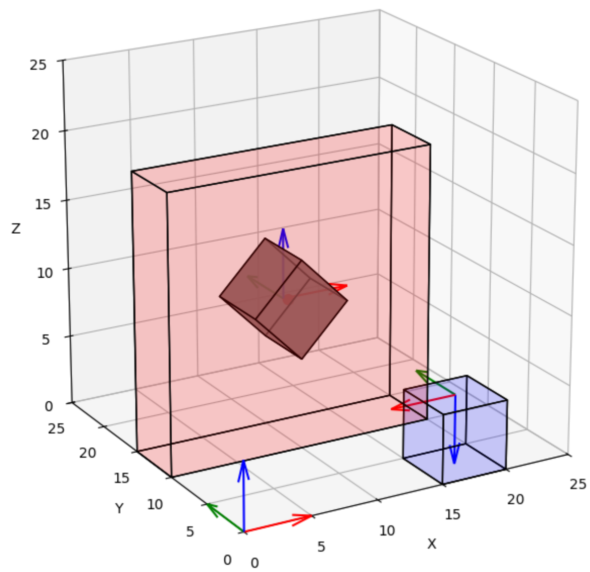

Figure 1: End-Effector Visual - Top Grasp Position

4 Task 4: Target Poses (Pre-Grasp & Final)

We calculate the cube's world-frame transformation matrix for two scenarios involving the rhombus-shaped hole in the red box. The hole is rotated by 45° around the Y-axis.

4.1 Part 1: Pre-Grasp Pose (Next to Hole)

The cube is aligned with the hole but positioned slightly in front of it (at $Y = 7.5$) to prepare for insertion.

- **Rotation:** $R_y(45^\circ)$
- **Translation:** $p = [10,\ 7.5,\ 10]^T$

$$T_{\text{pre-grasp}} \approx \begin{bmatrix} 0.707 & 0 & 0.707 & 10 \\ 0 & 1 & 0 & 7.5 \\ -0.707 & 0 & 0.707 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4.2 Part 2: Final Pose (Inside the Hole)

The cube is precisely positioned inside the box.

- **Rotation:** $R_y(45°)$
- **Translation:** $p = [10,\ 12.5,\ 10]^T$

$$T_{\text{final}} \approx \begin{bmatrix} 0.707 & 0 & 0.707 & 10 \\ 0 & 1 & 0 & 12.5 \\ -0.707 & 0 & 0.707 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4.3 Kinematic Model Parameters

Based on the kinematic structure of the Elephant myArm 300 Pi, we defined the geometric parameters required for the Product of Exponentials (PoE) formula. The dimensions were converted from millimeters to meters.

Link Lengths (Size) The vertical offsets between the joints are defined as:

$$d_1 = 0.1695\,\text{m}$$
$$d_3 = 0.1155\,\text{m}$$
$$d_5 = 0.1278\,\text{m}$$
$$d_7 = 0.0660\,\text{m}$$

Screw Axes (Twist Configuration) For each joint $i = 1 \dots 7$, we define the rotation axis unit vector ω_i and a point q_i on that axis relative to the base frame {S} in the zero configuration:

- **Joint 1:** Vertical rotation at base.

$$\omega_1 = [0, 0, 1]^T, \quad q_1 = [0, 0, 0]^T$$

- **Joint 2:** Horizontal rotation (shoulder).

$$\omega_2 = [0, 1, 0]^T, \quad q_2 = [0, 0, d_1]^T$$

- **Joint 3:** Vertical rotation.

$$\omega_3 = [0, 0, 1]^T, \quad q_3 = [0, 0, d_1 + d_3]^T$$

- **Joint 4:** Horizontal rotation (elbow).

$$\omega_4 = [1, 0, 0]^T, \quad q_4 = [0, 0, d_1 + d_3]^T$$

- **Joint 5:** Vertical rotation.

$$\omega_5 = [0, 0, 1]^T, \quad q_5 = [0, 0, d_1 + d_3 + d_5]^T$$

- **Joint 6:** Horizontal rotation (wrist).

$$\omega_6 = [1,0,0]^T, \quad q_6 = [0,0,d_1+d_3+d_5]^T$$

- **Joint 7:** Vertical rotation (end-effector flange).

$$\omega_7 = [0,0,1]^T, \quad q_7 = [0,0,d_1+d_3+d_5+d_7]^T$$

5 Task 5: Trajectory Design in $SE(3)$

5.1 Problem Definition

The goal is to design a smooth trajectory in $SE(3)$ that allows the manipulator to:

1. Approach the blue cube with the correct grasp pose (top grasp).
2. After grasping, move the cube to the "right next to the box's hole" pose (pre-grasp pose near the rhombus-shaped hole).

Both position and orientation must vary smoothly along the motion, so that the end-effector pose follows a continuous curve in $SE(3)$.

5.2 Key Poses in $SE(3)$

We define three main end-effector poses:

- **Home pose** T_s: initial robot pose above the workspace.
- **Top grasp pose** T_g: pose used to grasp the blue cube from the top.
- **Pre-hole pose** T_h: pose where the cube is aligned with the hole and located just in front of it.

In homogeneous coordinates (all positions in meters):

$$T_s = \begin{bmatrix} I_{3\times3} & \begin{bmatrix} 0 \\ 0 \\ 0.4788 \end{bmatrix} \\ 0_{1\times3} & 1 \end{bmatrix}.$$

For the top grasp, the end-effector is rotated by 180° around the X-axis (to point downwards) and positioned at the cube location:

$$R_g = R_x(\pi), \quad p_g = \begin{bmatrix} 0.175 \\ 0.025 \\ 0.05 \end{bmatrix}, \quad T_g = \begin{bmatrix} R_g & p_g \\ 0_{1\times3} & 1 \end{bmatrix}.$$

For the pose next to the box hole, the end-effector is rotated to align with the rhombus orientation and translated to a point in front of the hole:

$$R_h = R_y\left(\frac{\pi}{2}\right) R_x\left(\frac{\pi}{4}\right), \quad p_h = \begin{bmatrix} 0.10 \\ 0.125 \\ 0.10 \end{bmatrix}, \quad T_h = \begin{bmatrix} R_h & p_h \\ 0_{1\times3} & 1 \end{bmatrix}.$$

5.3 Implementation and Visualization

The trajectory is implemented in Python using NumPy, SciPy (`logm`, `expm`), and Matplotlib for 3D visualization. The code defines rotation matrices, homogeneous transformations, the cubic spline function in $SE(3)$, and a `draw_box` utility for rendering workspace objects.

Key visualization elements:

- Red box (target with rhombus-shaped hole, rotated 45° around Y).
- Blue cube (grasp target).
- Black trajectory line $\{p_k\}$.
- Markers for T_s (green), T_g (cyan), T_h (red).

This implementation ensures smooth motion without discontinuities in position or orientation, suitable for real robot execution.

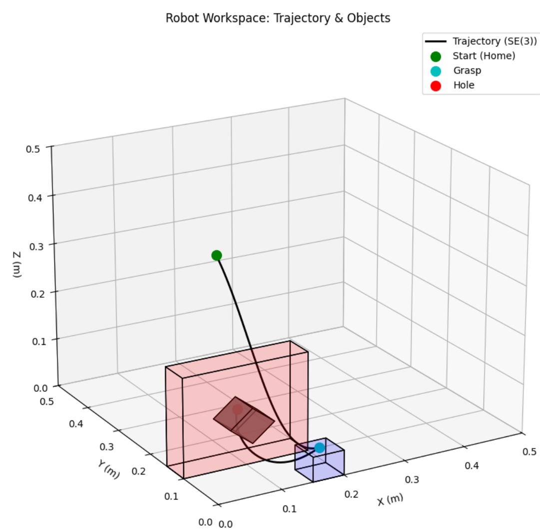

Figure 2: 3D workspace visualization showing the $SE(3)$ trajectory from home (green) to top grasp (cyan) to pre-hole pose (red), with objects rendered.

6 Kinematic Simulator for Pick-and-Place Task

The kinematic simulator implements Product of Exponentials (PoE) forward and inverse kinematics for a 7-DOF robotic arm. The simulation demonstrates a complete pick-and-place task using Jacobian pseudoinverse control in SE(3) task space with interactive GUI control.

6.1 Core Implementation Components

The algorithm leverages screw theory for precise kinematic modeling:

- **Screw Axes**: 7 screws ($S_i = [\omega_i, v_i]$) define joint motion axes
- **Forward Kinematics**: $T = e^{[\xi_1]\theta_1} \cdots e^{[\xi_7]\theta_7} M$
- **Geometric Jacobian**: $J = \frac{\partial T}{\partial \theta}$ (numerical differentiation)
- **Task-space Control**: $\dot{q} = J^{\dagger}(K \cdot e_{SE(3)})$

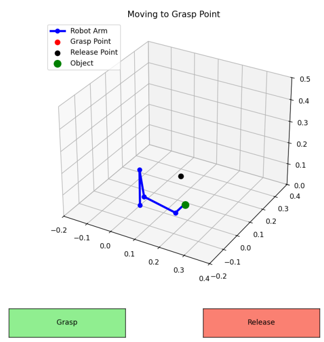

Figure 3: Pre-grasp phase. The arm approaches the object at $T_{grasp} = [0.175, 0.025, 0.05]^T$ using SE(3) trajectory control.

6.2 Interactive Control Interface

Figure 4: Interactive GUI buttons.

The control loop executes as follows:

1. Compute SE(3) error $e_{SE(3)} = [e_p, e_\omega]$ between current T_{curr} and goal T_{goal}
2. Jacobian pseudoinverse: $J^\dagger = J^T(JJ^T + \lambda I)^{-1}$ (damped least squares)
3. Integration: $q_{k+1} = q_k + \Delta t \dot{q}$

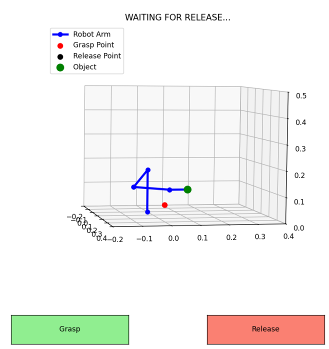

Figure 5: Release phase. After grasp, the object is transported to $T_{release} = [0.1, 0.125, 0.10]^T$ maintaining grasp orientation.

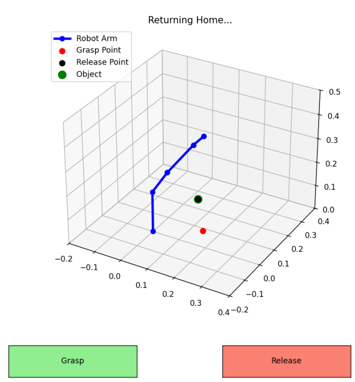

Figure 6: Return to home phase. Final linear interpolation in joint space: $q(t) = (1-\alpha)q_{final} + \alpha q_{home}$.

6.3 Simulation Output Matrices

During the simulation execution, the end-effector transformation matrices $T \in SE(3)$ at critical timestamps were recorded as follows:

Grasped object pose:

$$T_{grasp} = \begin{bmatrix} 8.66024384 \cdot 10^{-1} & -5.00001765 \cdot 10^{-1} & 2.72498441 \cdot 10^{-5} & 1.74649334 \cdot 10^{-1} \\ -3.30735026 \cdot 10^{-5} & -2.78522154 \cdot 10^{-6} & 9.99999999 \cdot 10^{-1} & 2.50546992 \cdot 10^{-2} \\ -5.00001765 \cdot 10^{-1} & -8.66024385 \cdot 10^{-1} & -1.89488794 \cdot 10^{-5} & 5.02133385 \cdot 10^{-2} \\ 0.00000000 & 0.00000000 & 0.00000000 & 1.00000000 \end{bmatrix}$$

Released object pose:

$$T_{release} = \begin{bmatrix} 8.66025399 \cdot 10^{-1} & -5.00000008 \cdot 10^{-1} & -2.45682306 \cdot 10^{-7} & 1.00003197 \cdot 10^{-1} \\ 2.48647276 \cdot 10^{-7} & -6.06948996 \cdot 10^{-8} & 1.00000000 & 1.25003633 \cdot 10^{-1} \\ -5.00000008 \cdot 10^{-1} & -8.66025399 \cdot 10^{-1} & 7.17603157 \cdot 10^{-8} & 9.99993612 \cdot 10^{-2} \\ 0.00000000 & 0.00000000 & 0.00000000 & 1.00000000 \end{bmatrix}$$